

Qingyang Zhang

stevenz3@andrew.cmu.edu | +1 412-657-6580 | linkedin.com/in/qingyazhang | qingyazhang.github.io

RESEARCH INTERESTS

Machine learning for healthcare; multimodal and longitudinal representation learning; medical imaging; generalization across clinical datasets.

EDUCATION

Carnegie Mellon University

August 2025 – May 2027 (expected)

M.S. in Information Networking · Pittsburgh, United States

Selected coursework: Machine Learning, Large Language Model Applications, Multimodal Machine Learning, Machine Learning Systems

New York University

August 2021 – May 2025

B.S. in Computer Science & Data Science · New York, United States · GPA: 3.85/4.0

Undergraduate capstone: Radiology Report Generation on Knee XR (2024).

Selected coursework: Data Structure, Algorithms, Computer Vision, Natural Language Processing, Deep Learning, Discrete Mathematics, Convex and Non-Smooth Optimization, Mathematical Statistics, Ordinary Differential Equation.

PUBLICATION

[1] **Qingyang Zhang**. Bridging the EHR Divide: Asymmetric Contrastive Learning for Cross-National Medical Representation Transfer.

NeurIPS 2026 GenAI4Health Workshop. Accepted (Poster).

[2] Y. Chen, H. Yang, H. Pan, F. Siddiqui, A. Verdone, **Qingyang Zhang**, S. Chopra, C. Zhao, and Y. Shen. BURExtract-Llama: An LLM for Clinical Concept Extraction in Breast Ultrasound Reports.

Proceedings of the 1st International Workshop on Multimedia Computing for Health and Medicine (MCHM), 2024, pp. 53–58.

RESEARCH EXPERIENCE

Longitudinal Multimodal Disease Prediction

Aug 2026 – Present

Carnegie Mellon University | Advisor: Min Xu

- Investigating whether prior admissions improve ICD-9 diagnosis prediction from clinical notes, radiology reports, chest X-rays, and vital-sign time series.
- Developing longitudinal patient-data integration and comparisons across history lengths using MIMIC-IV, MIMIC-IV-Note, and MIMIC-CXR.

MRI Reconstruction

May 2026 – Present

NYU Grossman School of Medicine | Advisor: Sumit Chopra

- Implemented and benchmarked four MRI reconstruction methods on the fastMRI brain multicoil dataset.
- Developing reconstruction comparisons and validating image-fidelity evaluation using NMSE, PSNR, and SSIM.

Cross-National EHR Representation Learning & Domain Adaptation

May 2025 – August 2026

Carnegie Mellon University | Advisor: Chenyan Xiong

- Implemented task-specific temporal Transformer encoders with asymmetric supervised contrastive learning, clustering trajectories sharing a positive outcome without explicitly attracting heterogeneous negative trajectories to one another.
- Built PySpark/Slurm processing for a 3.98-million-patient NHIRD cohort, constructing task-specific examples with up to five years of history, patient-level splits, and enforced prediction boundaries.
- Aligned seven clinical coding systems to NHIRD using direct mappings and Qwen3-Embedding-8B/FAISS retrieval; analyzed vocabulary coverage and mapping collisions across similarity thresholds.
- Developed a fine-tuning data-selection strategy using bag-of-words similarity to identify records relevant to target downstream tasks, retaining positive examples and selecting external negatives to control class ratios.
- In controlled objective ablations on a 400K-patient pre-training subset, achieved the highest mean AUPRC on 3 of 4 downstream tasks, including mortality AUPRC of 0.538 versus 0.517 for standard SupCon.
- Transferred NHIRD encoders to MIMIC-IV and EHRSHOT; at 16 target labels per class, exceeded the best published baselines on all five EHRSHOT incident-disease tasks (Acute MI AUPRC: 0.153 versus 0.118).

Automated Radiology Report Generation

May 2024 – Aug 2025

NYU Grossman School of Medicine | Advisors: Sumit Chopra and Yiqiu Shen

- Developed a two-stage knee X-ray reporting system that conditions GPT-4 on predicted pathology labels and a report template refined with physicians. Benchmarked CNN and Transformer models for knee X-ray pathology classification, achieving best macro-AUCs of 0.8999 on OAI (DenseNet) and 0.5910 on internal NYU Langone data (TResNet).
- Built a custom vision-language knee X-ray report-generation pipeline combining a Swin Transformer image encoder with Llama 3; achieved ROUGE-1 of 0.4625 and METEOR of 0.3740 on 593 internal test cases against standardized reference reports.
- Fine-tuned MAIRA-1 and MAIRA-2 on the internal knee X-ray dataset as a parallel approach to report generation.
- Curated approximately 3,000 paired image-report cases with GPT-4-assisted labels and manual review of 200 cases; implemented DICOM view selection, OCR text removal, and morphological image preprocessing.
- Received positive qualitative feedback from physicians on reports generated by the Swin-Llama 3 and MAIRA approaches.

Clinical NLP and Pathology Extraction

May 2024 – Aug 2024

NYU Grossman School of Medicine | Advisor: Sumit Chopra

- Designed and implemented a radiology-report pathology extraction system using GPT, pre-training and fine-tuning BERT models including NYUTron, and LoRA fine-tuning of Llama 2/3.
- Increased residents' exposure to clinically important pathologies by 135% through model-assisted case identification, broadening access to diverse cases to support medical training.

Chinese Terminology Extraction

Sep 2023 – Aug 2024

New York University | Advisor: Adam Meyers

- Led development of the Chinese version of Termolator, using keyword-guided Wikipedia retrieval to support domain-specific terminology extraction.
- Designed evaluation using independent encyclopedia sources to assess extracted terms in the absence of explicit ground-truth labels.
- Applied k-nearest-neighbor methods and graph-based label propagation to refine candidate terminology and glossary quality.

Domain Adaptation and Few-Shot Segmentation

Jan 2023 – Aug 2023

NYU Shanghai | Advisor: Li Guo

- Applied feature-similarity-based label propagation for domain adaptation and iterative label refinement, achieving 90% accuracy on the VisDA2017 benchmark.
- Investigated pretrained visual representations and segmentation architectures, including DINO, DeiT, PSPNet, UPerNet, and MaskFormer, on PASCAL VOC.

TEACHING EXPERIENCE

Teaching Assistant in Discrete Mathematics & Intro to Computer Science

Aug 2024 – May 2025

NYU Shanghai

- Provided eight hours of teaching and review sessions per week, including individual support for discrete mathematics, computer science and problem solving.
- Conducted an average of 28 tutoring appointments per month and received a 5/5 rating in tutor review surveys.

TECHNICAL SKILLS

Programming: Python, C++, R, SQL, Java

Machine learning: PyTorch, Transformers, scikit-learn, FAISS

Data and computing: PySpark, Pandas, NumPy, Slurm, Linux, Git, Docker